

Chambolle–Pock convergence with step-size product four

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Conjecture 00000008833

Both originals assert that Chambolle–Pock convergence always requires the product of the two step sizes to be at most one. No normalization of the coupling operator is imposed. The following actual scalar convex problem disproves this necessary-condition claim.

Problem and exact algorithm

Use the real Hilbert spaces $X = Y = \mathbb{R}$, and set

$$f(x) = \frac{1}{2}x^2, \quad g(z) = \frac{1}{2}z^2, \quad Kx = \frac{1}{4}x, \quad \tau = \sigma = 2, \quad \theta = 1.$$

Both objectives are finite, continuous, proper and convex. The coupling is nonzero and bounded, with $K^* = K$. Completing the square gives

$$g^*(y) = \sup_z \{yz - \frac{1}{2}z^2\} = \frac{1}{2}y^2, \quad yz - \frac{1}{2}z^2 = \frac{1}{2}y^2 - \frac{1}{2}(z - y)^2.$$

The supremum is attained at $z = y$. For either proximal objective at step size two, completing the square gives the unique minimizer

$$\text{prox}_{2f}(a) = \text{prox}_{2g^*}(a) = a/3, \quad \frac{1}{2}u^2 + \frac{1}{4}(u - a)^2 = \frac{3}{4}(u - a/3)^2 + \frac{1}{6}a^2.$$

The saddle function is $L(x, y) = x^2/2 + xy/4 - y^2/2$. The point $(0, 0)$ is a saddle because $L(0, y) \leq L(0, 0) \leq L(x, 0)$.

Algorithm 1 of Chambolle–Pock [1] updates

$$y' = \text{prox}_{\sigma g^*}(y + \sigma K\bar{x}), \quad x' = \text{prox}_{\tau f}(x - \tau K^*y'), \quad \bar{x}' = x' + \theta(x' - x).$$

Consequently this instance has the exact update

$$y' = (y + \bar{x}/2)/3, \quad x' = (x - y'/2)/3, \quad \bar{x}' = 2x' - x = -x/3 - y'/3.$$

The usual initialization $\bar{x}^0 = x^0$ is included below.

Theorem 1. *Every initial state $(x^0, y^0, \bar{x}^0)$ converges to $(0, 0, 0)$ under this iteration, although $\tau\sigma = 4 > 1$.*

Proof. Let $M = \max(|x|, |y|, |\bar{x}|)$. The update satisfies

$$|y'| \leq M/2, \quad |x'| \leq 5M/12 \leq M/2, \quad |\bar{x}'| \leq M/3 + M/6 = M/2.$$

Thus $M_n \leq 2^{-n}M_0 \rightarrow 0$. Every coordinate converges to zero, so the primal-dual pair converges to the exhibited saddle from every initial pair. The positive step sizes have product four, contradicting the claimed universal necessity of a product at most one. $\square$

Scope and the standard operator-norm condition

The counterexample concerns exactly the unqualified numerical product bound stated in both source versions. The operator K is nonzero, the objectives meet the standard convex assumptions, and a saddle exists. Convergence is proved for every initial state, rather than only a fixed point initialization or a specially selected orbit. The extrapolation parameter is the standard value $\theta = 1$.

The usual sufficient condition in Chambolle–Pock’s theorem is $\tau\sigma\|K\|^2 < 1$ [1]. Here $\|K\| = 1/4$, so that product is $1/4$. Our result is therefore consistent with the standard theorem. The source omits this operator-norm factor and makes its step-product condition universal; it does not assume $\|K\| = 1$. The direct proof above does not invoke the convergence theorem. Failure of this first conjunct suffices to disprove the conjunction; the separate preconditioning claim need not be resolved.

Formal objects and complete coverage

The Lean file defines the actual real quadratic **energy** and proves **Continuous** and **ConvexOn** properties. Since it is real-valued everywhere, its properness has no exceptional infinite values. The operator **coupling** is a genuine bounded **ContinuousLinearMap** on the real Hilbert line. Its scalar action, nonzeroness and adjoint identity against every pair of vectors are proved.

fenchel_conjugate computes the genuine **sSup** of the range $z \mapsto yz - f(z)$. A universal upper bound and an attaining point establish boundedness, existence and the value of this supremum. **prox_exact** proves an if-and-only-if characterization of all global minimizers of the actual proximal objective. Thus the proximal maps used by the algorithm are derived, including their uniqueness. **zero_saddle** checks the saddle inequalities for all x, y .

step implements both proximal updates, the bounded coupling operator and the extrapolation step with the stated parameters. The adjoint identity justifies using the same scalar coupling in the primal update. **step_formula** derives the displayed recurrence. The orbit is defined by actual function iteration, and **size** is the maximum of all three absolute coordinate values. **one_step** and **size_contracts** prove its contraction. **orbit_bound** proves the bound $2^{-n}M_0$ for every natural n . Finally, **orbit_converges** proves convergence in the ordinary product topology for every initial triple, and **product_bound_counterexample** combines that convergence with positive steps, nonzero coupling and $\tau\sigma > 1$.

Reproduction

Lean 4.19.0 and Mathlib commit

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are pinned. From `lean/`, run `lake update`, `lake exe cache get`, `lake build`, and `lake env lean Main.lean -DwarningAsError=true`. The six printed dependency audits use only standard logical axioms. No admitted statements or auxiliary computations are used.

References

- [1] A. Chambolle and T. Pock, *A first-order primal-dual algorithm for convex problems with applications to imaging*, J. Math. Imaging Vis. 40 (2011), 120–145. Algorithm 1 and Theorem 1. Author preprint.